Duhem's *Aim and Structure of Physical Theory* April 1, 2026

Background and Context

- Published as articles in 1904–05; first book edition 1906, second (with appendices) 1914
- Period of profound transformations in physics (electron, quantum hypothesis, relativity, nuclear atom). Much of the turmoil reflected efforts to reconcile the continuum approach to electromagnetism (Maxwell's equations, field theory) with the emerging atomistic picture of matter and energy. Duhem was an active participant in these debates, contributor to continuum mechanics, thermodynamics, etc.; favored treating matter using continuum mechanics.
- Duhem's stated aim: "a simple logical analysis of the method by which physical science makes progress" (p. 3), combined with articulation and defense of a particular view of how physics ought to proceed
- Part I (Chs. 1–4): the aim of physical theory; Part II (Chs. 4–10): structure (how theoretical operations contribute to progress)
- N.B.: Duhem's arguments rely heavily on detailed case studies, which makes the text both rich and challenging; furthermore, the positions he develops depend in subtle ways on then current debates in physics and applied mathematics (see Wilson, "Duhem before Breakfast")

Part I: The Aim of Physical Theory

Chapters 1–2: Explanation vs. Representation

The central question: what is the aim of a physical theory? Two broad alternatives (p. 7):

1. **Explanation**: aim is to explain experimentally established laws by describing the underlying reality responsible for them—to "strip reality of appearances covering it like a veil, in order to see bare reality itself"
2. **Representation and Classification**: aim is "to summarize and classify logically a group of experimental laws without claiming to explain these laws"

What "explanation" means: not "answers to why-questions" in the sense of later philosophy of science. Rather: to provide an as-if account of the underlying structure of objects and processes such that observed phenomena are consequences of that structure. Note that Duhem has no objection when we have genuine empirical access to posited processes (e.g., vibrational theory of sound); the objection targets *hypothetical* explanations, where the underlying reality "cannot be rendered accessible to the senses."

The autonomy argument against explanation as the aim:

- If the aim is to explain, physics must address questions about the nature of underlying reality
- The experimental method is "acquainted only with sensible appearances and can discover nothing beyond them" (p. 10)—these questions cannot be settled by the method of physics
- Physics therefore loses its *autonomy*: becomes subordinate to metaphysics

- Consequence: rather than universal consent, physics faces interminable disputes reflecting competing metaphysical schools

A more plausible (?) version:

- Given any as-if explanation, we can construct alternatives ("weakly equivalent" theories) that agree on all appearances but differ on claims about underlying reality
- Considerations of simplicity are *inappropriate* for choosing among these, because there is no reason underlying reality should conform to the simpler theory
- Therefore, the choice must rest on metaphysical grounds (e.g., what counts as a legitimate explanation, what types of entities are admissible)
- Result: adequacy of any theory depends on prior metaphysical commitments; physics cannot secure its results by its own methods alone

Scope of the threat: not merely that disputes *could* arise, but that they *have*—and that they concerned the very question of what constitutes a possible explanation ("occult causes" objections; Newton's forces at a distance as explanans or explanandum)

Assessment: argument not decisive, but shifts the burden of proof: advocates of explanation must show how the experimental method can suffice to choose among weakly equivalent theories. Duhem adopts a minimal-risk position: first secure a view under which physics is autonomous, then perhaps re-open the question of explanation.

Duhem's alternative: a physical theory is

> "a system of mathematical propositions, deduced from a small number of principles, which aim to represent as simply, as completely, and as exactly as possible a set of experimental laws." (p. 19)

Four operations in constructing such a theory:

1. Identification and measurement of physical magnitudes
2. Selection of hypotheses (restricted only by logic, not metaphysics)
3. Mathematical development: combining principles to enable calculations
4. Comparison of consequences with experiment

"Agreement with experiment is the sole criterion of truth for a physical theory" (p. 21).

Beyond instrumentalism: Duhem is *not* a simple instrumentalist (economy of thought, à la Mach). Theories also provide a *classification* of experimental laws; at their best, a *natural classification*:

> "The aim of physical theory is to become a natural classification, to establish among diverse experimental laws a logical coordination serving as a sort of image and reflection of the true order according to which the realities escaping us are organized." (p. 31)

Natural vs. artificial classification:

- Natural classification makes a claim (however limited) about underlying reality; divisions reflect real differences in nature
- Differs from explanation: leaves open what underlying structure is responsible
- Differs from instrumentalism: the question "Is this classification natural?" is substantive; given competing classifications, at most one is natural
- Evidential support: theory's ability to *project* beyond given laws, especially to *unanticipated* new laws subsequently confirmed
- In addition: strong evidence if classification combines simplicity, completeness, and exactness, which are in tension

Chapters 3–4: History, Models, and Progress

Central descriptive claim (Ch. 3): theories divide into two weakly linked parts:

- Abstract, mathematical, *representational* part: classifies experimental laws
- *Explanatory* part: purports to describe underlying mechanisms
- Representational part develops cumulatively; explanatory part shows no comparable growth
- "When the progress of experimental physics goes counter to a theory and compels it to be modified or transformed, the purely representative part enters nearly whole in the new theory … whereas the explanatory part falls out" (p. 32)

Challenge from model-based reasoning (Ch. 4): the English school (Faraday, Kelvin, Maxwell) relied on mechanical models to develop theories, most importantly, Maxwell's electromagnetic theory. Question: what role do the models play, and is it compatible with initial line of argument contrasting explanation and representation? Do models play a *constitutive* role in reasoning?

- Models represent, classify, and project to new laws, just as Duhemian theories do
- Duhem concedes heuristic value but denies models are part of the theory itself (or play constitutive role)
- Once formulated, equations stand independently (example: ether models subsequently discarded)
- Harder case: atomic/molecular models (Thomson $\rightarrow$ Bohr $\rightarrow$ quantum mechanics), where there may be a stronger case that models play constitutive role

Part II: Theory and Experiment

Chapter 6: Principles Governing the Comparison of Theory with Experiment

Background thesis: theoretical terms in physics (e.g., 'force,' 'temperature') are not simple referring expressions like 'swan' or 'green.' Two key features:

- **Symbolic**: theoretical statements express relations among symbols whose connection to the world is mediated through a complex web of provisional theories and idealizations. E.g., $T(x, y, z, t) = 42\,^{\circ}\mathrm{C}$ presupposes a theory of heat, a theory of thermometry, and idealizations about the body.

- **Approximate**: theoretical statements have no greater claim to truth than an indefinite number of other formulations that agree with observation to within the same range of accuracy.

From these two features, Duhem derives *three principles*:

Principle 1: Relative Falsification (the "Duhem Thesis")

- Because hypotheses express symbolic relationships, they can only be falsified *relative to* theoretical assumptions needed to give them empirical content
- These include: background assumptions within the same theory, background theories pertaining to measurement techniques, and idealizations of the empirical situation
- Schematically: if $H \,\&\, A_1 \,\&\, \cdots \,\&\, A_n$ entails (under conditions $C_1, \ldots, C_m$) a prediction P, and $\neg P$ is observed, then $\neg(H \,\&\, A_1 \,\&\, \cdots \,\&\, A_n)$—but *not* which element is false
- Individual hypotheses not falsified; *theories* are
- Significance: explains why testing in mature sciences involves extensive background assumptions—because the theses being tested are (partially interpreted) symbolic relationships

Principle 2: No Definitive Confirmation

- Because hypotheses express *approximations*, no "crucial experiment" can definitively establish a hypothesis, even with all background assumptions granted
- An indefinite number of hypotheses remain weakly equivalent on the same data—residual imprecision always leaves alternatives open
- However: experimentation *narrows* the spread among acceptable hypotheses—precision begets more precision
- Significance: explains why confirmation in mature sciences is tenuous, and why increasing precision of measurement drives theoretical progress

Principle 3: Holistic Comparison

- Because hypotheses are both approximate and symbolic: "the only experimental check on physical theory which is not illogical consists in comparing the entire system of the physical theory with the whole group of experimental laws" (p. 200)
- Cannot add hypotheses one at a time, establishing each via experiment—hypotheses cannot be isolated experimentally, and even if they could, data underdetermines the choice
- Evidence accumulates *holistically*: as parts of the theory are used in experiments and measurements, evidence accrues for the theory as a whole
- Significance: explains the dominant role of theories in mature sciences, in contrast to Baconian data-gathering picture

Important consequences:

- When experiment confutes a theory, *logic does not dictate where to revise*—any hypothesis can in principle be maintained by modifying others
- *Advocacy*—arguing that one revision is more promising on grounds of simplicity, completeness, exactness—is rational and necessary, not a sign of irrationality
- Every hypothesis in principle revisable; in practice, more central hypotheses better protected
- Distinctive pattern of progress: theories recognized as specifiable idealizations, with more refined theories reducing two forms of indeterminacy:
  - (i) in the physical interpretation of theoretical terms
  - (ii) in the fit between theoretical and empirical claims

Questions for Discussion

1. Duhem argues that explanation as the aim would subordinate physics to metaphysics. Is this persuasive? Can the experimental method suffice to choose among "weakly equivalent" theories; or does scientific practice always depend on background commitments about what counts as a good explanation?
2. Natural classification occupies a middle ground between instrumentalism and realism, in some sense. What exactly is the difference between a natural classification and a description of underlying structure? Is this a stable position?
3. Do the three principles (relative falsification, no definitive confirmation, holistic comparison) really follow from the symbolic and approximate character of physical theories? Or are there additional assumptions needed to establish the three principles?
4. Duhem (on p. 8, in the discussion of acoustics) grants that theories at least sometimes include claims that can make a claim to permanence – that is, they are not subject to being discarded later as science progresses. What conditions need to be met to make a case that underlying entities or processes have such a status?
5. In several passages (e.g. pp. 211, 216) Duhem suggests that some claims are more central to physical theories, and demand more evidence to be revised. How can Duhem account for the idea that evidence has differential bearing on different claims within a theory? (Is there any obstacle to extending his ideas to include such an account?)